

SOLIDWORKS API (.NET VBA C#) — Reference (130/337): IGtol (part 3 of 3)

SOLIDWORKS API Documentation (converted from CHM)

IGtol (part 3 of 3) (Type)

SOLIDWORKS API Help

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SetAllOver Method (IGtol)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGtol Interface](#) : SetAllOver Method (IGtol)

Sets whether this Gtol uses an All Over leader.

.NET Syntax

Visual Basic (Declaration)

```
Sub SetAllOver( _  
    ByVal AllOver As System.Boolean _  
)
```

Visual Basic (Usage)



```
Dim instance As IGtol  
Dim AllOver As System.Boolean  
  
instance.SetAllOver(AllOver)
```

C#

```
void SetAllOver(  
    System.bool AllOver  
)
```

C++/CLI

```
void SetAllOver(  
& System.bool AllOver  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

AllOver

True to use an All Over leader, false to not

Visual Basic for Applications (VBA) Syntax

See [Gtol::SetAllOver](#).

Remarks

This property is valid only for bent, perpendicular, and multi-jog leaders.

Use:

- [IGtol::IsAttached](#) to determine whether this symbol is currently using a leader.
- [IGtol::HasExtraLeader](#) to determine whether this symbol is using a bent leader.
- [IGtol::GetLeaderSide](#) to determine where the leader is attached to the symbol.
- [IGtol::SetLeader](#) to set the characteristics of the leader on this symbol.

See Also

[IGtol Interface](#)
[IGtol Members](#)
[IGtol::GetAllOver Method \(\)](#)
[IGtol::SetAllAroundThisSide Method \(\)](#)
[IGtol::SetAllOverThisSide Method \(\)](#)

Availability

SOLIDWORKS 2026 SP2, Revision Number 34.2

SetAllOverThisSide Method (IGtol)

[See Also](#)[SolidWorks.Interop.sldworks Namespace](#) > [IGtol Interface](#) : SetAllOverThisSide Method (IGtol)

Sets whether this Gtol uses an All Over This Side leader.

.NET Syntax

Visual Basic (Declaration)

```
Sub SetAllOverThisSide( _
    ByVal AllOverTS As System.Boolean _
)
```

Visual Basic (Usage)



```
Dim instance As IGtol
Dim AllOverTS As System.Boolean

instance.SetAllOverThisSide(AllOverTS)
```

C#

```
void SetAllOverThisSide(
    System.bool AllOverTS
)
```

C++/CLI

```
void SetAllOverThisSide(
    & System.bool AllOverTS
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

AllOverTS

True to use an All Over This Side leader, false to not

Visual Basic for Applications (VBA) Syntax

See [Gtol::SetAllOverThisSide](#).

Example

VBA Preconditions:

'Open a drawing.

```
Dim swApp As SldWorks
Dim Part As ModelDoc2
Dim myGtol As Gtol
Dim myAnno As Annotation
Dim boolstatus As Boolean
Dim longstatus As Long, longwarnings As Long
Option Explicit
Sub main()

    Set swApp = Application.SldWorks
    Set Part = swApp.ActiveDoc

    Set myGtol = Part.InsertGtol()
    If Not myGtol Is Nothing Then
        myGtol.SetFrameSymbols2 1, "<IGTOL-SPROF>", False, "", False, "", "", "", ""
        myGtol.SetFrameValues 1, ".031265", "", "", "", ""
        myGtol.SetFrameSymbols2 2, "", False, "", False, "", "", "", ""
        myGtol.SetFrameValues 2, "", "", "", "", ""
        myGtol.SetPTZHeight "", False
    End If
End Sub
```

```

myGtol.SetCompositeFrame False
myGtol.SetText 4, ""
myGtol.SetBetweenTwoPoints False, "", ""
myGtol.SetAllOverThisSide True
Set myAnno = myGtol.GetAnnotation()
If Not myAnno Is Nothing Then
    boolstatus = myAnno.SetPosition(0.801796955941255, 0.800162656875834, 0)
    longstatus = myAnno.SetLeader3(swLeaderStyle_e.swBENT, 0, True, False, False, False)
End If
End If
Part.WindowRedraw

```

End Sub

Remarks

This property is valid only for bent, perpendicular, and multi-jog leaders.

Use:

- [IGtol.IsAttached](#) to determine whether this symbol is currently using a leader.
- [IGtol.HasExtraLeader](#) to determine whether this symbol is using a bent leader.
- [IGtol.GetLeaderSide](#) to determine where the leader is attached to the symbol.
- [IGtol.SetLeader](#) to set the characteristics of the leader on this symbol.

See Also

[IGtol Interface](#)
[IGtol Members](#)
[IGtol::GetAllOverThisSide Method \(\)](#)
[IGtol::SetAllAroundThisSide Method \(\)](#)

Availability

SOLIDWORKS 2018 FCS, Revision Number 26.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

SetBelowFrameTextAt Method (IGtol)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGtol Interface](#) : SetBelowFrameTextAt Method (IGtol)

Edits or inserts a text line at the specified below frame line index of this GTol.

.NET Syntax

Visual Basic (Declaration)

```

Function SetBelowFrameTextAt( _
    ByVal Index As System.Integer, _
    ByVal Text As System.String, _
    ByVal Overwrite As System.Boolean _
) As System.Boolean

```

Visual Basic (Usage)



```

Dim instance As IGtol
Dim Index As System.Integer
Dim Text As System.String
Dim Overwrite As System.Boolean
Dim value As System.Boolean

```

```
value = instance.SetBelowFrameTextAt(Index, Text, Overwrite)
```

C#

```

System.bool SetBelowFrameTextAt(
    System.int Index,
    System.string Text,

```

```
        System.bool Overwrite  
    )
```

C++/CLI

```
System.bool SetBelowFrameTextAt(  
    & System.int Index,  
    & System.String^ Text,  
    & System.bool Overwrite  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Index

1-based line index at which to edit or insert text

Text

New text

Overwrite

True to overwrite the text at Index; false to insert a new line at Index

Return Value

True if the text is successfully edited or inserted, false if not

Visual Basic for Applications (VBA) Syntax

See [Gtol::SetBelowFrameTextAt](#).

Example

See **Set Text in Datum Tags and GTols** examples in [IGTol](#).

See Also

[IGtol Interface](#)

[IGtol Members](#)

[IGtol::DeleteBelowFrameTextAt Method \(\)](#)

[IGtol::GetBelowFrameTextAt Method \(\)](#)

[IGtol::GetBelowFrameTextLineCount Method \(\)](#)

[IGtol::InsertBelowFrameTextAt Method \(\)](#)

Availability

SOLIDWORKS 2018 FCS, Revision Number 26.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

SetBetweenTwoPoints Method (IGtol)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGtol Interface](#) : SetBetweenTwoPoints Method (IGtol)

Enables or disables the between two points symbol and its texts.

.NET Syntax

Visual Basic (Declaration)

```
Sub SetBetweenTwoPoints( _  
    ByVal Between As System.Boolean, _  
    ByVal TextFrom As System.String, _  
    ByVal TextTo As System.String _  
)
```

Visual Basic (Usage)



Copy Code

```
Dim instance As IGtol
Dim Between As System.Boolean
Dim TextFrom As System.String
Dim TextTo As System.String

instance.SetBetweenTwoPoints(Between, TextFrom, TextTo)
```

C#

```
void SetBetweenTwoPoints(
    System.bool Between,
    System.string TextFrom,
    System.string TextTo
)
```

C++/CLI

```
void SetBetweenTwoPoints(
    & System.bool Between,
    & System.String^ TextFrom,
    & System.String^ TextTo
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Between

True to enable the between two points symbol, or false to disable it

TextFrom

Text on the left end of the symbol

TextTo

Text on the right end of the symbol

Visual Basic for Applications (VBA) Syntax

See [Gtol::SetBetweenTwoPoints](#).

Example

[Insert Gtol \(C#\)](#)

[Insert Gtol \(VB.NET\)](#)

[Insert Gtol \(VBA\)](#)

Remarks

The TextFrom and TextTo values might be set to empty strings.

This method ignores TextFrom and TextTo if the between value is false (the symbol is being disabled).

Use:

- [IGtol::GetBetweenTwoPoints](#) to determine if this symbol is currently in use
- [IGtol::GetBetweenTwoPointsText](#) to determine the texts that are part of this symbol

See Also

[IGtol Interface](#)

[IGtol Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

SetCompositeFrame Method (IGtol)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGtol Interface](#) : SetCompositeFrame Method (IGtol)

Obsolete. Superseded by [IGtol::SetCompositeFrame2](#).

.NET Syntax

Visual Basic (Declaration)

```
Sub SetCompositeFrame( _  
    ByVal Composite As System.Boolean _  
)
```

Visual Basic (Usage)



```
Dim instance As IGtol  
Dim Composite As System.Boolean  
  
instance.SetCompositeFrame(Composite)
```

C#

```
void SetCompositeFrame(  
    System.bool Composite  
)
```

C++/CLI

```
void SetCompositeFrame(  
& System.bool Composite  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Composite

True to put the first two frames of this Gtol into a composite frame, false to not

Visual Basic for Applications (VBA) Syntax

See [Gtol::SetCompositeFrame](#).

Remarks

Sets whether to put the first two frames of this Gtol into a composite frame.

See Also

[IGtol Interface](#)
[IGtol Members](#)
[IGtol::GetCompositeFrame Method](#)

Availability

SOLIDWORKS 998Plus, datecode 1998319

SOLIDWORKS API Help

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SetCompositeFrame2 Method (IGtol)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGtol Interface](#) : SetCompositeFrame2 Method (IGtol)

Sets whether to create a composite frame containing the specified Gtol frame.

.NET Syntax

Visual Basic (Declaration)

```
Sub SetCompositeFrame2( _  
    ByVal Composite As System.Boolean, _  
    ByVal FrameNum As System.Short _  
)
```

Visual Basic (Usage)



```
Dim instance As IGtol
Dim Composite As System.Boolean
Dim FrameNum As System.Short

instance.SetCompositeFrame2(Composite, FrameNum)
```

C#

```
void SetCompositeFrame2(
    System.bool Composite,
    System.short FrameNum
)
```

C++/CLI

```
void SetCompositeFrame2(
    & System.bool Composite,
    & System.short FrameNum
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Composite
True to create a composite frame, false to not (see **Remarks**)
FrameNum
Index of GTol frame

Visual Basic for Applications (VBA) Syntax

See [Gtol::SetCompositeFrame2](#).

Example

[Create GTol Composite Frame \(VBA\)](#)
[Create GTol Composite Frame \(VB.NET\)](#)
[Create GTol Composite Frame \(C#\)](#)

Remarks

If Composite is true, this method creates a composite frame containing adjacent GTol frames:

- Frame with index FrameNum.
- Frame directly below.

Both GTol frames must have the same symbol.

See Also

[IGtol Interface](#)
[IGtol Members](#)
[IGtol::GetCompositeFrame2 Method \(\)](#)

Availability

SOLIDWORKS 2014 FCS, Revision Number 22.0

SOLIDWORKS API Help

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SetDatumIdentifier Method (IGtol)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGtol Interface](#) : SetDatumIdentifier Method (IGtol)

Sets the name of the datum being defined.

.NET Syntax

Visual Basic (Declaration)

```
Sub SetDatumIdentifier( _  
    ByVal DatumIdentifier As System.String _  
)
```

Visual Basic (Usage)



```
Dim instance As IGtol  
Dim DatumIdentifier As System.String  
  
instance.SetDatumIdentifier(DatumIdentifier)
```

C#

```
void SetDatumIdentifier(  
    System.string DatumIdentifier  
)
```

C++/CLI

```
void SetDatumIdentifier(  
& System.String^ DatumIdentifier  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

DatumIdentifier
Name of datum

Visual Basic for Applications (VBA) Syntax

See [Gtol::SetDatumIdentifier](#).

See Also

[IGtol Interface](#)
[IGtol Members](#)
[IGtol::GetDatumIdentifier Method](#)

SOLIDWORKS API Help

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SetDisplayDualDimensionInRangeValues Method (IGtol)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGtol Interface](#) : SetDisplayDualDimensionInRangeValues Method (IGtol)

Sets whether to display dual units in dimension range values.

.NET Syntax

Visual Basic (Declaration)

```
Sub SetDisplayDualDimensionInRangeValues( _  
    ByVal InValue As System.Boolean _  
)
```

Visual Basic (Usage)



```
Dim instance As IGtol  
Dim InValue As System.Boolean  
  
instance.SetDisplayDualDimensionInRangeValues(InValue)
```

C#

```
void SetDisplayDualDimensionInRangeValues(  
    System.bool InValue  
)
```

C++/CLI

```
void SetDisplayDualDimensionInRangeValues(
& System.bool InValue
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

InValue

True to display dual units in dimension range values, false to not

Visual Basic for Applications (VBA) Syntax

See [Gtol::SetDisplayDualDimensionInRangeValues](#).

See Also

[IGtol Interface](#)

[IGtol Members](#)

[IGtol::GetDisplayDualDimensionInRangeValues Method \(\)](#)

Availability

SOLIDWORKS 2024 FCS, Revision Number 32

SOLIDWORKS API Help

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SetFrameSymbols Method (IGtol)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGtol Interface](#) : SetFrameSymbols Method (IGtol)

Obsolete. Superseded by [IGtol::SetFrameSymbols2](#).

.NET Syntax

Visual Basic (Declaration)

```
Sub SetFrameSymbols( _
    ByVal FrameNumber As System.Short, _
    ByVal GCS As System.Short, _
    ByVal TolDia1 As System.Boolean, _
    ByVal TolMC1 As System.Short, _
    ByVal TolDia2 As System.Boolean, _
    ByVal TolMC2 As System.Short, _
    ByVal DatumMC1 As System.Short, _
    ByVal DatumMC2 As System.Short, _
    ByVal DatumMC3 As System.Short _
)
```

Visual Basic (Usage)



```
Dim instance As IGtol
Dim FrameNumber As System.Short
Dim GCS As System.Short
Dim TolDia1 As System.Boolean
Dim TolMC1 As System.Short
Dim TolDia2 As System.Boolean
Dim TolMC2 As System.Short
Dim DatumMC1 As System.Short
Dim DatumMC2 As System.Short
Dim DatumMC3 As System.Short
```

```
instance.SetFrameSymbols(FrameNumber, GCS, TolDia1, TolMC1, TolDia2, TolMC2, DatumMC1, DatumMC2, DatumMC3)
```

C#

```
void SetFrameSymbols(
    System.short FrameNumber,
```

```
System.short GCS,  
System.bool ToIDia1,  
System.short ToIMC1,  
System.bool ToIDia2,  
System.short ToIMC2,  
System.short DatumMC1,  
System.short DatumMC2,  
System.short DatumMC3  
)
```

C++/CLI

```
void SetFrameSymbols(  
& System.short FrameNumber,  
& System.short GCS,  
& System.bool ToIDia1,  
& System.short ToIMC1,  
& System.bool ToIDia2,  
& System.short ToIMC2,  
& System.short DatumMC1,  
& System.short DatumMC2,  
& System.short DatumMC3  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

FrameNumber

GCS

ToIDia1

ToIMC1

ToIDia2

ToIMC2

DatumMC1

DatumMC2

DatumMC3

Visual Basic for Applications (VBA) Syntax

See [Gtol::SetFrameSymbols](#).

See Also

[IGtol Interface](#)
[IGtol Members](#)

SOLIDWORKS API Help

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SetFrameSymbols2 Method (IGtol)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGtol Interface](#) : SetFrameSymbols2 Method (IGtol)

Sets the symbols for the specified frame.

.NET Syntax

Visual Basic (Declaration)

```
Sub SetFrameSymbols2( _  
    ByVal FrameNumber As System.Short, _  
    ByVal GCS As System.String, _  
    ByVal ToIDia1 As System.Boolean, _  
    ByVal ToIMC1 As System.String, _  
    ByVal ToIDia2 As System.Boolean, _  
    ByVal ToIMC2 As System.String, _  
    ByVal DatumMC1 As System.String, _
```

```

    ByVal DatumMC2 As System.String, _
    ByVal DatumMC3 As System.String _
)

```

Visual Basic (Usage)



```

Dim instance As IGtol
Dim FrameNumber As System.Short
Dim GCS As System.String
Dim TolDia1 As System.Boolean
Dim TolMC1 As System.String
Dim TolDia2 As System.Boolean
Dim TolMC2 As System.String
Dim DatumMC1 As System.String
Dim DatumMC2 As System.String
Dim DatumMC3 As System.String

instance.SetFrameSymbols2(FrameNumber, GCS, TolDia1, TolMC1, TolDia2, TolMC2, DatumMC1, DatumMC2, DatumMC3)

```

C#

```

void SetFrameSymbols2(
    System.short FrameNumber,
    System.string GCS,
    System.bool TolDia1,
    System.string TolMC1,
    System.bool TolDia2,
    System.string TolMC2,
    System.string DatumMC1,
    System.string DatumMC2,
    System.string DatumMC3
)

```

C++/CLI

```

void SetFrameSymbols2(
& System.short FrameNumber,
& System.String^ GCS,
& System.bool TolDia1,
& System.String^ TolMC1,
& System.bool TolDia2,
& System.String^ TolMC2,
& System.String^ DatumMC1,
& System.String^ DatumMC2,
& System.String^ DatumMC3
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

FrameNumber

Feature control frame 1 for first

GCS

Geometric tolerance symbol (see **Remarks**)

TolDia1

Diameter symbol exists for tolerance 1 (True or false)

TolMC1

Material condition symbol for tolerance 1 (see **Remarks**)

TolDia2

Diameter symbol exists for tolerance 2 (True or false)

TolMC2

Material condition symbol for tolerance 2 (see **Remarks**)

DatumMC1

Material condition symbols for primary datum (see **Remarks**)

DatumMC2

Material condition symbols for secondary datum (see **Remarks**)

DatumMC3

Material condition symbols for tertiary datum (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [Gtol::SetFrameSymbols2](#).

Example

[Insert GTol \(C#\)](#)
[Insert GTol \(VB.NET\)](#)
[Insert GTol \(VBA\)](#)

 Remarks

This method is valid only if this GTol was created in a version of SOLIDWORKS earlier than 2022.

GTol symbols are defined in **C:\ProgramData\SolidWorks\SolidWorks 20nn\langlenglish\gtol.sym**. Reference this file to specify GCS, TolMC1, TolMC2, DatumMC1, DatumMC2, and DatumMC3 in <LibraryName-SymbolName> format.

Examples:

- GCS = "<IGTOL-POS>" (Position symbol in the ISO Geometric Tolerancing Symbols library)
- TolMC1 = "<MOD-LMC>" (Least Material Condition symbol in the Modifying Symbols library)

After calling this method, call [IGtol::SetFrameValues2](#) to specify tolerance and datum values.

 See Also

[IGtol Interface](#)
[IGtol Members](#)
[IGtol::GetFrameCount Method](#)
[IGtol::GetFrameDiameterSymbols Method](#)
[IGtol::GetFrameSymbols2 Method](#)
[IGtol::GetFrameValues Method](#)
[IGtol::IGetFrameDiameterSymbols Method](#)
[IGtol::IGetFrameSymbols2 Method](#)
[IGtol::IGetFrameValues Method](#)

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SetFrameValues Method (IGtol)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGtol Interface](#) : SetFrameValues Method (IGtol)

Obsolete. Superseded by [IGtol::SetFrameValues2](#).

 .NET Syntax

Visual Basic (Declaration)

```
Sub SetFrameValues( _  
    ByVal FrameNumber As System.Short, _  
    ByVal Tol1 As System.String, _  
    ByVal Tol2 As System.String, _  
    ByVal Datum1 As System.String, _  
    ByVal Datum2 As System.String, _  
    ByVal Datum3 As System.String _  
)
```

Visual Basic (Usage)



```
Dim instance As IGtol  
Dim FrameNumber As System.Short  
Dim Tol1 As System.String  
Dim Tol2 As System.String  
Dim Datum1 As System.String  
Dim Datum2 As System.String  
Dim Datum3 As System.String  
  
instance.SetFrameValues(FrameNumber, Tol1, Tol2, Datum1, Datum2, Datum3)
```

C#

```
void SetFrameValues(  
    System.short FrameNumber,  
    System.string Tol1,  
    System.string Tol2,  
    System.string Datum1,  
    System.string Datum2,  
    System.string Datum3  
)
```

C++/CLI

```

void SetFrameValues(
& System.Short FrameNumber,
& System.String^ Tol1,
& System.String^ Tol2,
& System.String^ Datum1,
& System.String^ Datum2,
& System.String^ Datum3
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

FrameNumber

Feature control frame 1 for first

Tol1

Tolerance 1 value

Tol2

Tolerance 2 value

Datum1

Datum reference 1 (primary datum)

Datum2

Datum reference 2 (secondary datum)

Datum3

Datum reference 3 (tertiary datum)

Visual Basic for Applications (VBA) Syntax

See [Gtol::SetFrameValues](#).

See Also

[IGtol Interface](#)

[IGtol Members](#)

[IGtol::GetFrameCount Method](#)

[IGtol::GetFrameDiameterSymbols Method](#)

[IGtol::GetFrameSymbols2 Method](#)

[IGtol::GetFrameValues Method](#)

[IGtol::IGetFrameSymbols2 Method](#)

[IGtol::IGetFrameValues Method](#)

[IGtol::SetFrameSymbols2 Method](#)

SOLIDWORKS API Help

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SetFrameValues2 Method (IGtol)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGtol Interface](#) : SetFrameValues2 Method (IGtol)

Sets the geometric tolerance values and datum references in the specified frame of this GTol symbol.

.NET Syntax

Visual Basic (Declaration)

```

Function SetFrameValues2( _
    ByVal FrameNumber As System.Short, _
    ByVal Tol1 As System.String, _
    ByVal Tol2 As System.String, _
    ByVal Datum1 As System.String, _
    ByVal Datum2 As System.String, _
    ByVal Datum3 As System.String _
) As System.Boolean

```

Visual Basic (Usage)

```

Dim instance As IGtol
Dim FrameNumber As System.Short
Dim Tol1 As System.String
Dim Tol2 As System.String
Dim Datum1 As System.String
Dim Datum2 As System.String

```



```
Dim Datum3 As System.String
Dim value As System.Boolean
```

```
value = instance.SetFrameValues2(FrameNumber, Tol1, Tol2, Datum1, Datum2, Datum3)
```

C#

```
System.bool SetFrameValues2(
    System.short FrameNumber,
    System.string Tol1,
    System.string Tol2,
    System.string Datum1,
    System.string Datum2,
    System.string Datum3
)
```

C++/CLI

```
System.bool SetFrameValues2(
    & System.short FrameNumber,
    & System.String^ Tol1,
    & System.String^ Tol2,
    & System.String^ Datum1,
    & System.String^ Datum2,
    & System.String^ Datum3
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

FrameNumber

Symbol frame number; 1 for first frame (see **Remarks**)

Tol1

Tolerance 1 value

Tol2

Tolerance 2 value

Datum1

Primary datum with material conditions (see **Remarks**)

Datum2

Secondary datum with material conditions (see **Remarks**)

Datum3

Tertiary datum with material conditions (see **Remarks**)

Return Value

True if frame values are set, false if not

Visual Basic for Applications (VBA) Syntax

See [Gtol::SetFrameValues2](#).

Example

[Insert GTol \(VBA\)](#)

[Insert GTol \(VB.NET\)](#)

[Insert GTol \(C#\)](#)

Remarks

This method is valid only if this GTol was created in a version of SOLIDWORKS earlier than 2022.

Before calling this method, call:

- [IGtol::SetFrameSymbols2](#) to specify the GTol symbol for this frame.
- [IGtol::GetFrameCount](#) to specify FrameNumber.

Geometric tolerance and material condition symbols are defined in **C:\ProgramData\SolidWorks\SolidWorks 20nn\langlenglishgtol.sym**. Inspect this file as you specify Datum1, Datum2, and Datum3 in <LibraryName-SymbolName> format.

For example, for datum tags A, B, and C:

```
bRet = aGTol.SetFrameValues2(1, "0.2", "", "B-A-C<MOD-MMC>", "B<MOD-MMC>-C<MOD-LMC>", "C<MOD-MMC>-A")
```

sets the first symbol frame of this GTol with:

- Tolerance 1 = 0.2
- Primary datum = B-A-C<MOD-MMC>
- Secondary datum = B<MOD-MMC>-C<MOD-LMC>
- Tertiary datum = C<MOD-MMC>-A

See Also

[IGtol Interface](#)
[IGtol Members](#)
[IGtol::GetFrameDiameterSymbols Method \(\)](#)
[IGtol::GetFrameSymbols2 Method \(\)](#)
[IGtol::GetFrameValues Method \(\)](#)

Availability

SOLIDWORKS 2016 FCS, Revision Number 24.0

SOLIDWORKS API Help

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SetFromToText Method (IGtol)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGtol Interface](#) : SetFromToText Method (IGtol)

Adds a From-To symbol and sets the From-To text for this Gtol.

.NET Syntax

Visual Basic (Declaration)

```
Function SetFromToText( _  
    ByVal EnableFromTo As System.Boolean, _  
    ByVal FromText As System.String, _  
    ByVal ToText As System.String _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IGtol  
Dim EnableFromTo As System.Boolean  
Dim FromText As System.String  
Dim ToText As System.String  
Dim value As System.Boolean  
  
value = instance.SetFromToText(EnableFromTo, FromText, ToText)
```

C#

```
System.bool SetFromToText(  
    System.bool EnableFromTo,  
    System.string FromText,  
    System.string ToText  
)
```

C++/CLI

```
System.bool SetFromToText(  
& System.bool EnableFromTo,  
& System.String^ FromText,  
& System.String^ ToText  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

EnableFromTo
True to enable From-To symbol
FromText
From text label
ToText
To text label

Return Value

True if From-To text successfully set, false if not

Visual Basic for Applications (VBA) Syntax

See [Gtol::SetFromToText](#).

Example

See the [IGtolFrame](#) examples.

Remarks

The From-To symbol:

- is present below the Gtol frame.
- specifies that the Gtol value applies from one point or entity to another.

See Also

[IGtol Interface](#)
[IGtol Members](#)
[IGtol::GetFromToText Method \(\)](#)

Availability

SOLIDWORKS 2023 FCS, Revision Number 31

SOLIDWORKS API Help

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SetLeader Method (IGtol)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGtol Interface](#) : SetLeader Method (IGtol)

Sets the characteristics of the leader for this symbol.

.NET Syntax

Visual Basic (Declaration)

```
Sub SetLeader( _  
    ByVal Leader As System.Boolean, _  
    ByVal LeaderSide As System.Integer, _  
    ByVal BentLeader As System.Boolean, _  
    ByVal AllAround As System.Boolean _  
)
```

Visual Basic (Usage)



```
Dim instance As IGtol  
Dim Leader As System.Boolean  
Dim LeaderSide As System.Integer  
Dim BentLeader As System.Boolean  
Dim AllAround As System.Boolean  
  
instance.SetLeader(Leader, LeaderSide, BentLeader, AllAround)
```

C#

```
void SetLeader(  
    System.bool Leader,  
    System.int LeaderSide,  
    System.bool BentLeader,  
    System.bool AllAround  
)
```

C++/CLI

```
void SetLeader(  
& System.bool Leader,  
& System.int LeaderSide,  
& System.bool BentLeader,  
& System.bool AllAround  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Leader

True enables a leader on this symbol, false disables it

LeaderSide

Leader attachment information as defined in [swLeaderSide_e](#)

BentLeader

True enables a bent leader on this symbol, false disables it

AllAround

True enables the all around symbol on the leader, false disables it

Visual Basic for Applications (VBA) Syntax

See [Gtol::SetLeader](#).

Remarks

This method ignores:

- LeaderSide, BentLeader, and AllAround values if the leader value is false. Use [IGtol::IsAttached](#) to determine if this symbol is currently using a leader.
- AllAround value if the BentLeader value is false. Use [IGtol::HasExtraLeader](#) to determine if this symbol is using a bent leader.

Use:

- [IGtol::GetLeaderSide](#) to determine where the leader is attached to the symbol
- [IGtol::GetAllAround](#) to determine if this leader is using the all around symbol

See Also

[IGtol Interface](#)

[IGtol Members](#)

[IGtol::GetLeaderAtIndex2 Method](#)

[IGtol::GetLeaderCount Method](#)

[IGtol::GetLeaderInfo Method](#)

[IGtol::IGetLeaderAtIndex2 Method](#)

[IGtol::IGetLeaderInfo Method](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

SetPosition Method (IGtol)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGtol Interface](#) : SetPosition Method (IGtol)

Sets the position of this GTol.

.NET Syntax

Visual Basic (Declaration)

```
Sub SetPosition( _  
    ByVal X As System.Double, _  
    ByVal Y As System.Double, _  
    ByVal Z As System.Double _  
)
```

Visual Basic (Usage)

```
Dim instance As IGtol  
Dim X As System.Double  
Dim Y As System.Double  
Dim Z As System.Double  
  
instance.SetPosition(X, Y, Z)
```

C#

```
void SetPosition(  
    System.double X,  
    System.double Y,
```



```
        System.double Z
    )
```

C++/CLI

```
void SetPosition(
&    System.double X,
&    System.double Y,
&    System.double Z
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

X
X location for the GTol

Y
Y location of the GTol

Z
Z location of the GTol

Visual Basic for Applications (VBA) Syntax

See [Gtol::SetPosition](#).

Example

The position is the upper-left corner of the GTol.

Example

[Get Gtol Witness Line \(VBA\)](#)
[Get Gtol Witness Line \(VB.NET\)](#)
[Get Gtol Witness Line \(C#\)](#)

See Also

[IGtol Interface](#)
[IGtol Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

SetPTZHeight Method (IGtol)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGtol Interface](#) : SetPTZHeight Method (IGtol)

Obsolete. Superseded by [IGtol::SetPTZHeight2](#).

.NET Syntax

Visual Basic (Declaration)

```
Sub SetPTZHeight( _
    ByVal PtzHt As System.String, _
    ByVal DisplayIn As System.Boolean _
)
```

Visual Basic (Usage)



```
Dim instance As IGtol
Dim PtzHt As System.String
Dim DisplayIn As System.Boolean

instance.SetPTZHeight(PtzHt, DisplayIn)
```

C#

```
void SetPTZHeight(
    System.string PtzHt,
```

```
        System.bool DisplayIn
    )
```

C++/CLI

```
void SetPTZHeight(
&    System.String^ PtzHt,
&    System.bool DisplayIn
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

PtzHt

Height of the projected tolerance zone

DisplayIn

True to display the projected zone tolerance, false to not

Visual Basic for Applications (VBA) Syntax

See [Gtol::SetPTZHeight](#).

Remarks

The projected tolerance zone (PTZ) displays in the first tolerance window of the first control frame of the GTol. If PtzHt is not empty, its value is displayed after the PTZ symbol, which is a P enclosed in a circle.

See Also

[IGtol Interface](#)

[IGtol Members](#)

[IGtol::GetPTZHeight Method](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

SetPTZHeight2 Method (IGtol)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGtol Interface](#) : SetPTZHeight2 Method (IGtol)

Sets the projected tolerance zone for the specified frame and tolerance in this GTol.

.NET Syntax

Visual Basic (Declaration)

```
Function SetPTZHeight2( _
    ByVal FrameNumber As System.Integer, _
    ByVal TolNumber As System.Integer, _
    ByVal PtzDisplay As System.Boolean, _
    ByVal PtzHt As System.String _
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IGtol
Dim FrameNumber As System.Integer
Dim TolNumber As System.Integer
Dim PtzDisplay As System.Boolean
Dim PtzHt As System.String
Dim value As System.Boolean
```

```
value = instance.SetPTZHeight2(FrameNumber, TolNumber, PtzDisplay, PtzHt)
```

C#

```
System.bool SetPTZHeight2(
    System.int FrameNumber,
    System.int TolNumber,
```

```

        System.bool PtzDisplay,
        System.string PtzHt
    )

```

C++/CLI

```

System.bool SetPTZHeight2(
& System.int FrameNumber,
& System.int TolNumber,
& System.bool PtzDisplay,
& System.String^ PtzHt
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

FrameNumber

Frame number

TolNumber

Tolerance number

PtzDisplay

True to display the projected zone tolerance, false to not

PtzHt

Height of the projected tolerance zone

Return Value

True if this method executed successfully, false if not

Visual Basic for Applications (VBA) Syntax

See [Gtol::SetPTZHeight2](#).

Example

[Insert Gtol \(C#\)](#)

[Insert Gtol \(VB.NET\)](#)

[Insert Gtol \(VBA\)](#)

Remarks

This method is valid only if this Gtol was created in a version of SOLIDWORKS earlier than 2022.

The projected tolerance zone (PTZ) displays in the first tolerance window of the first control frame of the Gtol. If PtzHt is not empty, its value is displayed after the PTZ symbol, which is a P enclosed in a circle.

See Also

[IGtol Interface](#)

[IGtol Members](#)

[IGtol::GetPTZHeight2 Method \(\)](#)

Availability

SOLIDWORKS 2017 FCS, Revision Number 25.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

SetText Method (IGtol)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGtol Interface](#) : SetText Method (IGtol)

Sets the specified text part of this Gtol.

.NET Syntax

Visual Basic (Declaration)

```

Function SetText( _
    ByVal WhichText As System.Integer, _

```

```
ByVal Text As System.String _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IGtol  
Dim WhichText As System.Integer  
Dim Text As System.String  
Dim value As System.Boolean  
  
value = instance.SetText(WhichText, Text)
```

C#

```
System.bool SetText(  
    System.int WhichText,  
    System.string Text  
)
```

C++/CLI

```
System.bool SetText(  
& System.int WhichText,  
& System.String^ Text  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

WhichText

Text to set as defined in [swGTolTextParts_e](#)

Text

New content for the specified WhichText

Return Value

True if successful, false if not

Visual Basic for Applications (VBA) Syntax

See [Gtol::SetText](#).

Example

[Set Text in Datum Tags and GTols \(VBA\)](#)

[Set Text in Datum Tags and GTols \(VB.NET\)](#)

[Set Text in Datum Tags and GTols \(C#\)](#)

Remarks

After calling this method, call [IModelView::GraphicsRedraw](#) to see the new text.

See Also

[IGtol Interface](#)

[IGtol Members](#)

[IGtol::GetText Method \(\)](#)

Availability

SOLIDWORKS 2011 SP03, Revision Number 19.3

SOLIDWORKS API Help

[Send comments on this topic.](#)

SetTextFormat Method (IGtol)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IGtol Interface](#) : SetTextFormat Method (IGtol)

Obsolete. Superseded by [IAnnotation::SetTextFormat](#).

.NET Syntax

Visual Basic (Declaration)

```
Function SetTextFormat( _  
    ByVal TextFormatType As System.Integer, _  
    ByVal TextFormat As System.Object _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IGtol  
Dim TextFormatType As System.Integer  
Dim TextFormat As System.Object  
Dim value As System.Boolean  
  
value = instance.SetTextFormat(TextFormatType, TextFormat)
```

C#

```
System.bool SetTextFormat(  
    System.int TextFormatType,  
    System.object TextFormat  
)
```

C++/CLI

```
System.bool SetTextFormat(  
& System.int TextFormatType,  
& System.Object^ TextFormat  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

TextFormatType

TextFormat

Visual Basic for Applications (VBA) Syntax

See [Gtol::SetTextFormat](#).

See Also

[IGtol Interface](#)
[IGtol Members](#)